



Dynamics of culling risk of Holstein cows in US organic dairy herds

Bhuwan Shrestha,¹ Pablo J. Pinedo,² Rajesh Neupane,¹ Robert Valeris-Chacin,³
Juan M. Piñeiro,¹ and Sushil Paudyal^{1*}

¹Department of Animal Science, Texas A&M University, College Station, TX 77843

²Department of Animal Sciences, Colorado State University, Fort Collins, CO 80523

³Texas A&M College of Veterinary Medicine & Biomedical Sciences, Canyon, TX 79015

ABSTRACT

The study aimed to analyze the culling dynamics of cows in US organic dairy herds based DHI-reported reasons. In addition, the overall cow longevity in these herds was evaluated. A total of 287,927 lactation records from cows calving between 2018 and 2022 across 434 organic dairy herds from 31 US states were evaluated. Annualized rates of live culling and death were calculated, and the distribution of culling reasons was assessed by parity, days after calving, season, year, and geographic region. The average age at culling and its association with lifetime genetic merit were also examined. The average age at culling was 62.4 mo. Reproduction was the leading reason for culling (27.0%), followed by died (13.2%), mastitis (12.7%), and low production (12.1%). Overall, the annualized culling rate was 28.4%, with 24.7% due to live culling and 3.7% associated with died. Seasonal variation was observed in culling due to reproduction, with a decline during the winter and an increase after summer. Early lactation was a high-risk period for culling reasons died, disease, and other udder problem, and late lactation was associated with elevated risk of culling due to reproduction, low production, feet and legs, injury/other, and mastitis. Cow-level factors such as calving season, parity, and cow-relative 305-d mature equivalent (305ME) milk yield were significantly associated with risk of dying within 60 d after calving. The risk of died within 60 d after calving was greater for cows calving in the fall compared with those calving in winter, spring, or summer, and higher for primiparous cows than for cows in parity ≥ 4 . Herd-level variables, including herd-relative 305-ME milk yield and herd size, were significantly associated with the risk of dying within 60 d postpartum. These same factors, along with the calving season, were associated with the risk of live culling during the first 60 d postpartum. High-producing cows based on

high cow-relative 305ME milk yield were more likely to be culled for feet and legs, reproduction, and died, whereas low-producing cows had higher odds of being culled for low production, mastitis, and udder problems. A moderate negative correlation was detected between age at culling and genetic merit indices. These findings highlight that culling dynamics in US organic dairy herds are influenced by lactation stage, parity, milk production, herd-level characteristics, and calving season.

Key words: culling, Holstein, organic dairy

INTRODUCTION

Culling is a fundamental practice within dairy management that involves the systematic removal of cows from the herd due to various voluntary or involuntary reasons. Voluntary culling involves the removal of cows from a herd due to factors such as low milk production, aggressive behavior, or being sold to other farms for dairy purposes. In contrast, involuntary culling is associated with factors such as disease, injury, suboptimal fertility, or mortality (Ahlman et al., 2011). Cows are less likely to be culled if they achieve timely conception, maintain good health, avoid occurrence of mastitis, exhibit efficient milk production, and possess sound udders, feet, and legs (De Vries and Marcondes, 2020). Culling is a critical strategy for enhancing the productivity and profitability of the herd by identifying and removing unproductive or diseased cows. However, the practice of culling too frequently or too hastily may not be economically viable for the dairy operation. Patterns of culling and the reasons cows leave the herd provide valuable insight into underlying herd-level challenges. Therefore, understanding how and why cows are removed is essential for developing effective management strategies and could improve herd management and support efforts to optimize herd longevity.

In the United States, approximately 35% to 38% of dairy cows are culled annually and replaced by newly raised heifers (Rotz et al., 2021). According to the De Vries (2020), the culling rate among Holstein cows is

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*Corresponding author: sushil.paudyal@ag.tamu.edu

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

34%, excluding those sold for dairy purposes, with cows typically remaining in production for fewer than 3 lactations, averaging just 35.3 mo. In contrast, organic dairy farms tend to have lower culling rates, with reported rates of 21% in Canada (Rozzi et al., 2007).

An earlier report by Bascom and Young (1998) identified reproductive failure, low production, and mastitis as leading causes of culling in conventional herds in New Hampshire, Vermont, and Massachusetts. Hadley et al. (2006) reported that from 1993 to 1999, the most frequent reasons for culling across 10 states of the Northeast and Upper Midwest region of the United States included injury/other, reproduction problems, low production, mastitis, and mortality, with an average annual cull rate of 31.6% in Holstein herds. Pinedo et al. (2010) found that “died” was the primary culling reason in Holstein conventional herds across 38 US states between 2001 and 2006, followed by reproduction, injury/other, and low production, with an overall culling rate of 30.7%. In 2014, low productivity was the main disposal code followed by died, mastitis, and injury for Jersey, Holstein, and Jersey $\times$ Holstein crossbred cows in a large conventional farm in Texas with an overall culling rate of 33.6% based on the records from 2007 to 2011 (Pinedo et al., 2014). However, the reasons for culling cows and the culling rate in organic dairy herds in the United States have not been extensively studied as compared with conventional dairy systems.

In Canadian organic herds, Rozzi et al. (2007) found that fertility issues accounted for 40% of culling cases in organic herds, followed by mastitis, feet and legs, milk production, and old age. In Swedish organic herds, poor udder health was the most common reason for culling, whereas low fertility ranked highest in conventional herds (Ahlman et al., 2011). Within the organic herds, Fernandes et al. (2021) reported that the cows with subclinical mastitis in the first month of lactation had a higher risk of died or culling during the first 300 d of lactation compared with cows without subclinical mastitis. Collectively, these findings demonstrate that culling reasons vary across production systems and countries; however, comprehensive information on culling dynamics in US organic dairy herds remains limited.

The National Organic Program (NOP) of USDA formulates the rules and guidelines related to organic livestock and products in the United States (USDA-AMS-NOP, 2011). According to these regulations, livestock must be fed and cared for using organic agricultural products. Organic dairy cows must have access to organic pasture for a minimum of 120 d each year and should receive at least 30% of their DMI from the pasture during the grazing season. Organic producers are restricted from using antibiotics and hormones, including reproductive hormones for the management of estrus and ovula-

tion (except for oxytocin for postparturition care), and parasiticides (except for fenbendazole and moxidectin). Animals lose organic status if unauthorized medications, such as antibiotics, are administered to them. As a result, organic dairy farming relies heavily on preventive strategies and natural remedies (Block, 2021). For some chronic diseases, like mastitis, culling remains one of the few viable options to protect the herd (Fernandes et al., 2021). These regulatory constraints make organic dairy production distinct from conventional systems and create unique challenges in animal health management, which may influence both culling rates and the reasons cows leave the herd. Therefore, the objective of this study was to analyze the culling dynamics of cows in US organic dairy herds based on DHI-reported reasons and to evaluate the overall cow longevity in these herds.

MATERIALS AND METHODS

Study Population

This study examined herd DHI lactation records from 434 organic dairy herds across 31 US states, consisting of a total of 287,927 records of cow calving between January 2018 and December 2022 (Supplemental Table S1, see Notes). Organic dairies were identified based on the price of milk received by the farm. The dataset was provided by Dairy Records Management Systems (DRMS, Raleigh, NC) and included cow- and herd-level information such as birth date, calving date, parity, date of dry-off, 305-d mature equivalent (305-ME) milk yield, herd code and state, breeding records, and lifetime genetic merit indices (net merit, cheese merit, fluid merit, graze merit), along with the recorded reason and date of culling.

Reasons for cow culling were reported as 10 different culling codes, which included: feet and legs, dairy purpose, low production, reproductive problems, injury/other, died, mastitis, disease, udder problems, and reason not reported (Norman et al., 2021). Each cow was assigned a singular (only one) culling code. Udder problems represent culling reasons aside from mastitis, such as poor udder conformation, udder injuries, or teat damage. The code “dairy purpose” was not considered a culling event because the cow was sold alive to another organic dairy with the expressed goal of continuing to provide income, such as producing milk or calves with continuation of her life cycle (Fetrow et al., 2006). Live culling included all disposal codes except died, and the code “died” was analyzed separately to calculate annualized death culling rates.

Data Handling and Management

The individual cow-related data within each farm including birth date, calving date, parity, date of dry-

off, 305-ME milk yield, herd code and state, breeding records, and lifetime genetic merit indices (net merit, cheese merit, fluid merit, graze merit) were obtained by combining separate lactation, breeding, identification, and genetics datasets received from DRMS. Lifetime genetic merit indices ranks dairy animals based on their combined genetic merit for economically important traits and are used to predict their lifetime profit (Cole et al., 2021). Net merit dollars are tailored to the average US pricing situation, cheese merit dollars emphasize milk protein for cheese production, fluid merit dollars place less emphasis on protein for fluid milk markets, and grazing merit dollars are designed for pasture-based herds using intensive grazing systems (Norman et al., 2022; CDCB, 2025b). Lactation records from herds averaging fewer than 10 calvings per year were excluded from the dataset. The culling date was identified as the last known reported status date with the code culling. Cow-days for each individual were calculated as the number of days from the first (or earliest) calving date to the culling or died date. Age at culling was determined by calculating the number of days from birth to the culling date. Herd-level parameters were derived from the merged lactation records by using information from cows within the herd. Herd milk yield (305ME) per year per season and herd size per year were calculated. Annualized live culling and death culling rates were computed using the equation (number of cows that were live culled or died/number of cow-days at risk) $\times$ 365 (Fetrow et al., 2006).

Events of Interest and Independent Variables

The event of interest in this study was culling due to 8 of the 10 recorded culling reasons, excluding “dairy purpose” and “reason not reported.” Categorical variables were created to assess live culling and death culling by 60 DIM because a substantial portion of these events typically occurs during early lactation (Dechow et al., 2012). Independent variables included cow-relative 305ME milk yield, herd-relative 305ME milk yield, herd size, and the season of culling.

All independent variables were categorized before analysis. The distribution of continuous variables was examined using histograms and descriptive statistics to guide the categorization process. Cow and herd 305ME milk yield were categorized as less than mean $-$ SD, mean $-$ SD to mean $+$ SD, and greater than mean $+$ SD. Cow-relative 305ME milk yield was defined as the deviation from the herd mates’ herd year-season average milk yield, and was categorized as: $<-1,944$ kg (low), $-1,944$ to $1,805$ kg (intermediate), and $>1,805$ kg (high). Herd-relative 305ME milk yield, calculated as the deviation from the herd’s year-season average (Pinedo et al., 2010; Pinedo et al., 2014), was catego-

rized as: $<-2,118$ kg (low), $-2,118$ to $1,823$ kg (intermediate), and $>1,823$ kg (high). Herd size was categorized based on the number of cows calving per year: ≤ 100 (small) and >100 (large). Parity was grouped as 1, 2, 3, and ≥ 4 . Both calving and culling seasons were defined as winter (January to March), spring (April to June), summer (July to September), and fall (October to December), allowing all seasons to be completed within the same year (Pinedo et al., 2014; Ferreira et al., 2020; Pattamanont et al., 2021).

Statistical Analysis

Annualized live culling and died rates were computed across various categories of explanatory variables (cow-relative 305ME milk yield, herd-relative 305ME milk yield, herd size, calving season, culling season, and parity). The distribution of reasons for disposal codes was examined by days after calving, season, and geographic regions. The geographic regions—northeast (Pennsylvania, New York, New Jersey), south (Louisiana, Mississippi, Texas), and west (California, Oregon, Washington)—were selected based on expected regional differences associated with climate.

Logistic regression (PROC GLIMMIX; SAS Institute Inc., Cary, NC) was used to assess the risk of death culling and live culling within 60 DIM, as well as for various disposal codes across categories of explanatory variables. Models for live and death culling were fitted separately to assess the effect of cow- and herd-related factors. Univariate analyses were first conducted to determine which explanatory variables to include in the models, with variables showing a P -value ≤ 0.20 considered for multivariable analysis. Final models were selected using a backward elimination approach, with herd included as the random variable.

Hazard distribution for specific disposal codes by parity was estimated using the Kaplan-Meier methods of the LIFETEST procedure of SAS. Calving date was considered the origin of time in the estimation of hazard functions, and the cow surviving until the subsequent calving was censored at that point. The records ending with the culling date were censored at the culling date. Records missing a subsequent calving date or a culling event and date were considered censored at the last known event date. The hazard analysis was conducted for the period from 1 to 500 d after calving. New variables were created for each reason for culling, where a value of 1 was assigned if a cow was culled for that specific reason, and 0 if the cow was removed for any other reason. Additionally, the average age at culling was calculated, and its correlation with lifetime genetic merit indices, including net merit, cheese merit, fluid merit, and grazing merit was assessed using the PROC CORR procedure in SAS.

RESULTS AND DISCUSSION

Distribution of Reason-Specific Culling

After edits, 287,927 lactations in 483 herds, ranging from 10 to 10,835 cows calving per year during the study period, were included in the analysis. Overall, 6% of the culling records were coded as “reason not reported” and 8% were coded as “dairy purpose.”

We identified an overall annualized culling rate of 28.4%, which consisted of 24.7% live culling and 3.7% death culling (Table 1). This rate in organic dairy farms is lower than the rates reported in conventional dairy farms. For instance, for cows calving between 2001 and 2006, Pinedo et al. (2010) reported an overall annualized culling rate of 31.7% with 25.1% live culling and 6.6% death culling. Later on, for cows calving between 2007 and 2011, the overall culling rate reported was 33.6% with 27.7% live culling and 5.9% death culling (Pinedo et al., 2014). Similarly, De Vries and Marcondes (2020) reported an average annual culling rate of 34% in 2019. The lower annualized cull rates observed in the current study may be associated with differences in production system and the economic situation during the years included in the analysis leading to altered culling decisions because the cow cull value was lowest during 2019–2020 among the 14 years (2010–2023; Portner and Heins, 2024), most likely due to an effect of COVID-19 pandemic (Martinez et al., 2021).

Annualized culling rate was lower for high-yielding cows compared with low-yielding cows; however, cows with intermediate milk yield experienced a higher culling rate than both low and high producers (Table 1). This finding is consistent with previous studies that reported lower culling rates for high-producing cows in conventional dairy farms (Miller et al., 2008; Pinedo et al., 2010; Pinedo et al., 2014). Similarly, Hadley et al. (2006) observed that high-producing cows were 0.5% to 1.7% less likely to be culled compared with the average producers. Milk yield is valued more in organic dairy farms because of the higher price of milk as well as the higher cost of production, leading to dairy farmers preferring to keep the high milk-producing animals for better economic returns. Lower culling and death rates observed in high-yielding cows may result from dairy producers providing preferential care and management to these cows over their lower-yielding counterparts (Dematawewa and Berger, 1998; Shahid et al., 2015). Alternatively, cows producing less milk may already be compromised by underlying health disorders or subclinical disease, which both limit milk production and increase their likelihood of removal, resulting in an apparent protective association between high milk yield and survivability.

Table 1. Annualized culling rate by subgroups of variables of interest

Subgroup (%)	Annualized rate ¹		
	Overall	Live culling	Death culling
All culling	28.4	24.7	3.7
Cow-relative 305ME milk yield			
<−1,944	20.9	19.3	1.6
−1,944 to 1,805	65.6	58.9	6.7
>1,805	10.8	9.6	1.2
Herd-relative 305ME milk yield			
<−2,118	19.2	17.7	1.5
−2,118 to 1,823	67.3	60.4	6.9
>1,823	10.9	9.7	1.2
Herd size			
≤100	7.7	6.7	1.0
>100	20.8	18.0	2.8
Calving season			
Winter	20.8	17.9	2.9
Spring	20.4	17.8	2.6
Summer	25.7	22.3	3.4
Fall	24.7	21.4	3.3
Culling season			
Winter	23.9	20.7	3.2
Spring	20.9	18.0	2.9
Summer	23.7	20.4	3.3
Fall	23.3	20.4	2.9
Parity			
1	30.4	26.6	3.8
2	28.1	24.6	3.5
3	24.5	21.2	3.3
≥4	42.5	36.5	6.0

¹Annualized live culling and death culling = [number of cows that were live culled (all codes except dairy purpose) and death (code died)/total number of cow-days at risk (all cows cow-days at risk in herd including dairy purpose)] × 365.

In terms of herd-relative 305ME milk yield, lower annualized culling rates were observed in herds with higher relative production, while herds with intermediate and low relative yields showed higher annualized culling rates than high-yielding herds. This pattern may reflect producers' tendency to replace lower and intermediate-producing cows with genetically superior heifers, thereby accelerating herd genetic progress (De Vries, 2017). Comparable patterns of low culling in high-production herds have been documented in previous research in conventional dairy herds (Smith et al., 2000; Pinedo et al., 2014).

Our results indicate that large herds had higher annualized culling rates than the small herds (Table 1). This aligns with earlier findings that showed a positive relationship between herd size and the risk of cow culling and mortality (Smith et al., 2000; Hadley et al., 2006; McConnel et al., 2008; Pinedo et al., 2010). Limited individual attention and delayed identification of at-risk cows for timely treatment may contribute to higher mortality and culling rates in larger herds (Shahid et al., 2015). Another reason for the high culling rate in large herds may be that producers adopt a higher herd replacement rate to capitalize on the genetic gains of newly improved heifers because larger herds are better able to absorb the

Table 2. Frequency distribution of culling with various DHI-reported disposal codes (as a % of total culling risk) for different subgroups

Subgroup (%)	Feet and legs	Low production	Reproduction	Injury/ other	Died	Mastitis	Disease	Udder	Reason not reported
All culling	5.9	12.1	27.0	11.2	13.2	12.7	8.3	3.2	6.1
Cow-relative 305ME milk yield									
<-1,944	4.9	20.9	24.2	9.1	7.5	17.1	7.6	3.6	5.1
-1,944 to 1,805	6.3	11.1	34.2	11.3	10.2	11.7	5.5	3.2	6.4
>1,805	6.1	9.6	43.9	10.3	10.9	6.5	5.4	2.1	5.2
Herd-relative 305ME milk yield									
<-2,118	4.5	22.5	23.8	9.1	7.6	17.0	7.2	3.6	4.8
-2,118 to 1,823	6.3	10.8	34.2	11.3	10.2	11.9	5.6	3.2	6.6
>1,823	6.7	10.3	43.2	9.8	10.7	6.7	6.0	2.2	4.5
Herd size									
≤100	5.6	11.0	18.9	13.6	12.7	16.3	3.0	2.9	16.1
>100	6.1	12.6	30.1	10.3	13.5	11.5	10.3	3.3	2.4
Calving season									
Winter	5.2	12.7	25.4	11.9	13.9	12.7	8.1	2.8	7.3
Spring	6.1	11.8	26.0	11.6	13.0	12.8	9.0	3.4	6.4
Summer	6.4	11.7	29.4	10.6	12.9	12.4	8.2	3.4	5.2
Fall	6.1	12.4	27.1	10.9	13.3	13.2	8.1	3.2	5.8
Culling season									
Winter	5.0	12.5	29.4	9.8	13.2	12.9	8.1	3.1	6.1
Spring	6.1	13.0	24.1	10.9	13.6	13.2	9.0	3.7	6.4
Summer	7.3	11.5	24.0	13.1	13.7	13.1	8.0	3.6	5.8
Fall	5.5	11.6	30.6	11.0	12.6	11.9	8.2	2.5	6.1
Parity									
1	4.3	14.0	29.0	11.1	12.6	10.4	9.3	2.1	7.3
2	5.0	12.3	30.8	12.0	12.3	11.1	8.6	1.7	6.3
3	5.6	11.1	28.1	12.2	13.6	13.6	7.8	2.6	5.6
≥4	8.0	11.4	22.7	10.2	14.2	15.1	7.7	5.3	5.4

replacement costs due to their greater financial resources (Almeida et al., 2025).

We identified the highest live culling (22.3%) and death culling rate (3.4%) for cows that calved during the summer (Table 1). Similarly, Dechow and Goodling (2008) also reported that calving during the summer season had a more negative effect on cow survival. Organic dairy cows are prone to seasonal variations because they need to walk several distances for pasture during grazing periods. In addition, heat stress associated with high temperature and humidity during the summer season for cows in their postpartum transition period may be associated with the increased death rate and culling of cows calving during summer season (Vitali et al., 2009). The largest annualized culling rate occurred during winter (23.9%) and summer (23.7%) months in these herds (Table 1), which could be due to the extreme cold and hot in winter and summer season, respectively. However, in conventional dairy farms, Pinedo et al. (2010) reported the largest culling rate during the spring season.

In our dataset, cows in parity ≥4 were at greater risk of culling, with 40%, 51%, and 73% higher risk compared with those in parities 1, 2, and 3 respectively (Table 1). Previous studies by Miller et al. (2008); Pinedo et al. (2014); Thomsen and Houe (2023) have reported that cows in higher parity were at greater risk for overall cull-

ing in conventional herds. Notably in this study, cows in parity 1 showed elevated annualized rates of both live and death culling compared with those in parities 2 and 3. A similar pattern of higher live culling among first-parity cows was observed by Thomsen and Houe (2023), whereas Pinedo et al. (2010) found that culling rates tended to increase progressively with parity. The elevated culling rate in primiparous cows compared with older cows may be associated with their higher susceptibility to calving difficulties (Olson et al., 2009; Morek-Kopeć et al., 2021), udder edema, and removal due to behavioral issues. Such difficulties often lead to a greater likelihood of forced culling because they negatively affect milk, fat, and protein production; extend days open; increase the number of services required; and result in greater cow losses (Morek-Kopeć et al., 2021). This result may also be related to unique management challenges in organic production systems during the heifer stage and first lactation, including greater reliance on pasture-based nutrition (Pinedo and Velez, 2019), challenges in achieving adequate growth rates and body condition at calving (Hadfield, 2020), the risk of severe negative energy balance in cows in early lactation due to the relatively low proportion of concentrate feed when following organic legislation (Flaten and Lien, 2009), all of which may lead to higher removal of primiparous animals. Therefore,

targeted management strategies for primiparous cows are required in organic dairies to help reduce their risk of culling, given the high cost of raising organic-certified dairy replacement heifers.

The most frequently reported reason for culling was reproduction, accounting for 27.0% of all culling events, followed by died (13.2%), mastitis (13.1%), low production (12.1%), and injury/other (11.2%; Table 2). These findings align with the study of Rozzi et al. (2007), who reported reproduction problem as the main reason for culling in organic dairy farms in Canada. Similarly, previous studies also identified reproduction as the leading reason for culling in dairy cows in conventional herds (Bascom and Young, 1998; Ahlman et al., 2011; Workie et al., 2021). Although Ahlman et al. (2011) found udder health to be the primary reason for culling in Swedish organic herds, this could be attributed to the geographic variation and European standards being different for organic dairy management.

The current results partially agree with the study by Pinedo et al. (2014), which reported “died” and “mastitis” as the second and third most common culling reasons in conventional herds, with low production as a leading reason of culling. Similarly, the DHIA disposal codes summary by the Council on Dairy Cattle Breeding, which includes both conventional and organic dairy information, listed low production as the leading reason, followed by died, reproduction, mastitis, and feet and legs (Norman et al., 2021). Additionally, 2 studies in New York also identified low production as the primary reason for culling in conventional herds, followed by reproduction and udder problems (Van Vleck and Norman, 1972; Milian-Suazo et al., 1988). However, in the present study, reproduction was the most frequently reported culling reason across all subcategories, including cow-relative and herd-relative 305ME milk yield, herd size, calving season, culling season, and parity. Because organic dairies are not permitted to use hormones of estrus or ovulation synchronization protocols for reproductive management, that may lead to a higher number of cows failing to become pregnant, ultimately resulting in their culling from the herd (Pinedo and Velez, 2019). Moreover, organic dairy farmers are reported to rely less on veterinarians and use veterinary services sparingly, including limited pregnancy diagnosis. This can delay the detection of pregnant or open cows, leading to negatively affecting reproductive efficiency and increasing the potential for cows being culled (Stiglbauer et al., 2013; Pinedo and Velez, 2019).

The distribution of risk of culling with various disposal codes relative to days after calving is presented in Figure 1. The relative frequency of culling reasons varied throughout the lactation period, with particularly notable changes observed for the categories “died” and “reproduction.” In

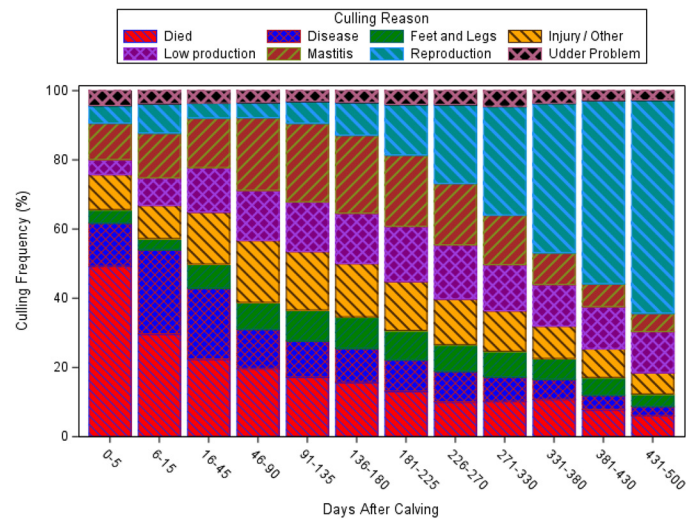


Figure 1. Distribution (%) of cows by culling reason across DIM (each day is 100%).

the first 90 d postcalving, “died” was the primary reason for culling, with the proportion decreasing from 50% to approximately 20%. This higher incidence of death in early lactation identified in our study aligns with findings from previous studies (Hadley et al., 2006; Dechow and Goodling, 2008; Pinedo et al., 2014). Dechow and Goodling (2008) concluded that herds with high early-lactation cull rates are likely to have poor transition cow health. The transition period is the critical period for survival of cows, where a shift from positive energy balance to negative energy balance occurs, leading to substantial immune suppression and thus increasing the risk of metabolic disease and risk of removal from the herd. (Hammon et al., 2006; Vergara et al., 2014; Horst et al., 2021). Notably, Figure 1 shows that during the 0 to 90 d postpartum window, “disease” was the second most frequent disposal code. LeBlanc et al. (2006) also reported around 75% of disease in dairy cows occurs in transition period. In organic production systems, this elevated disease risk poses a unique challenge; limited treatment options compared with conventional herds may further increase the likelihood of these disease cows eventually dying, as reflected in our early lactation mortality rates.

Reproduction was the main reported culling reason in late lactation (beyond 270 d after calving). The proportion of cows culled with disposal code reproduction increased from 5% during the first 90 d to 60% of all culling by 500 d after calving. The increased culling due to the reason reproduction at advanced lactation suggests cows were removed from the herd due to failure to conceive, likely being diagnosed as open during pregnancy checks at dry-off (Pinedo et al., 2014). This could also be attributed to the restrictions on using reproductive hormones or

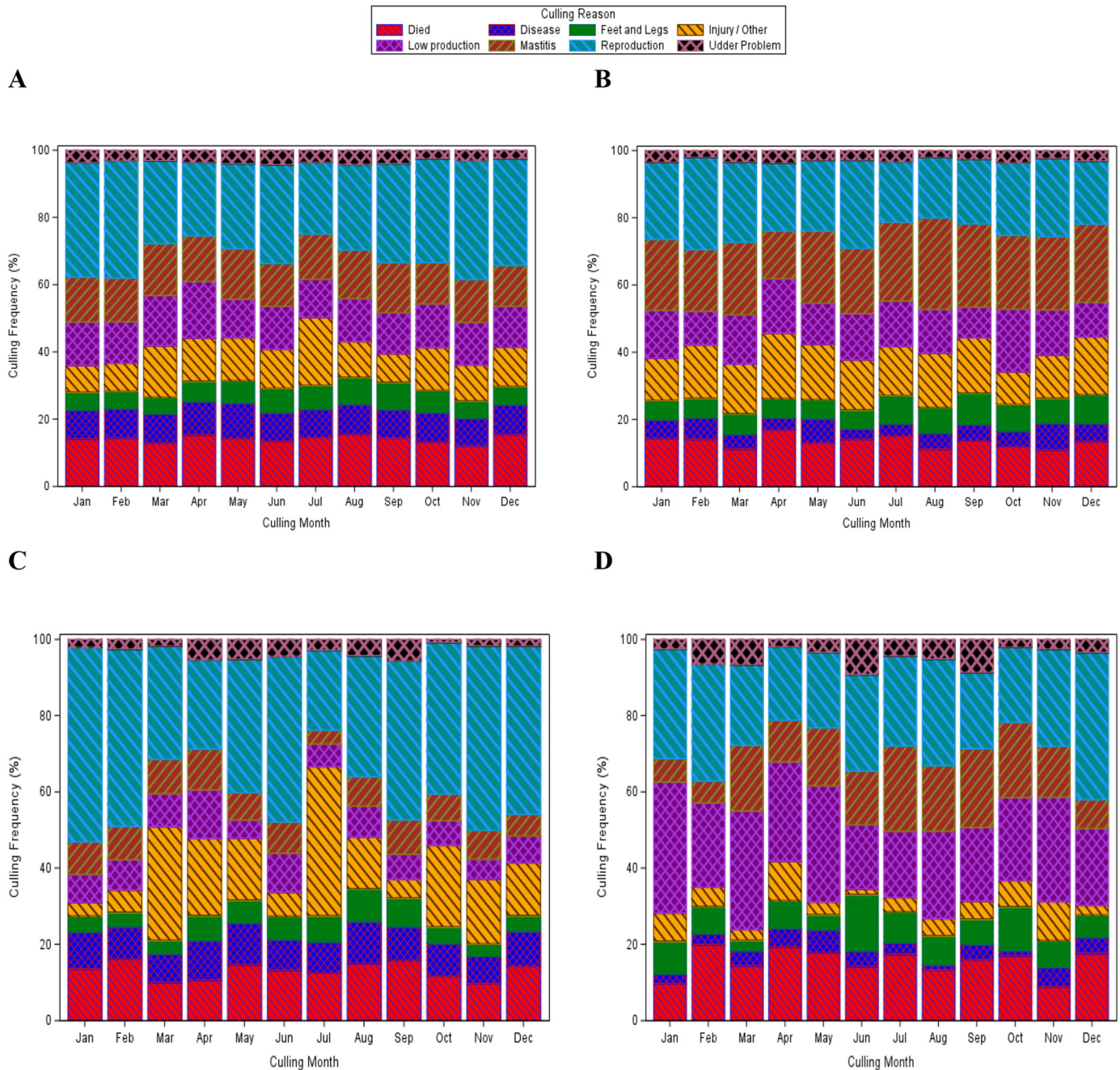


Figure 2. Distribution (%) of culling reasons by month of culling and geographic area: (A) overall, (B) northeast (Pennsylvania, New York, New Jersey), (C) south (Louisiana, Mississippi, Texas), and (D) west (California, Oregon, Washington). Each month adds up to 100%.

ovulation synchronization protocols by organic dairies (Pinedo and Velez, 2019). In our study, “mastitis” was the primary disposal code reported from 46 d to 225 d after calving, representing approximately 23% of all culling reasons. In conventional farms, Pinedo et al. (2010) reported mastitis as the primary disposal code from 150 d to 225 d after calving, representing 20% of all culling reasons. These results suggest that culling due to mastitis

was more prominent in early lactation in organic dairies, whereas it was more common during midlactation in conventional systems. This difference may be attributed to the limited treatment options for mastitis available in organic dairies, where the use of antibiotics is restricted. As a result, early-lactation culling rates tend to be higher. In contrast, conventional farms can often delay culling decisions for mastitis by utilizing available treatments.

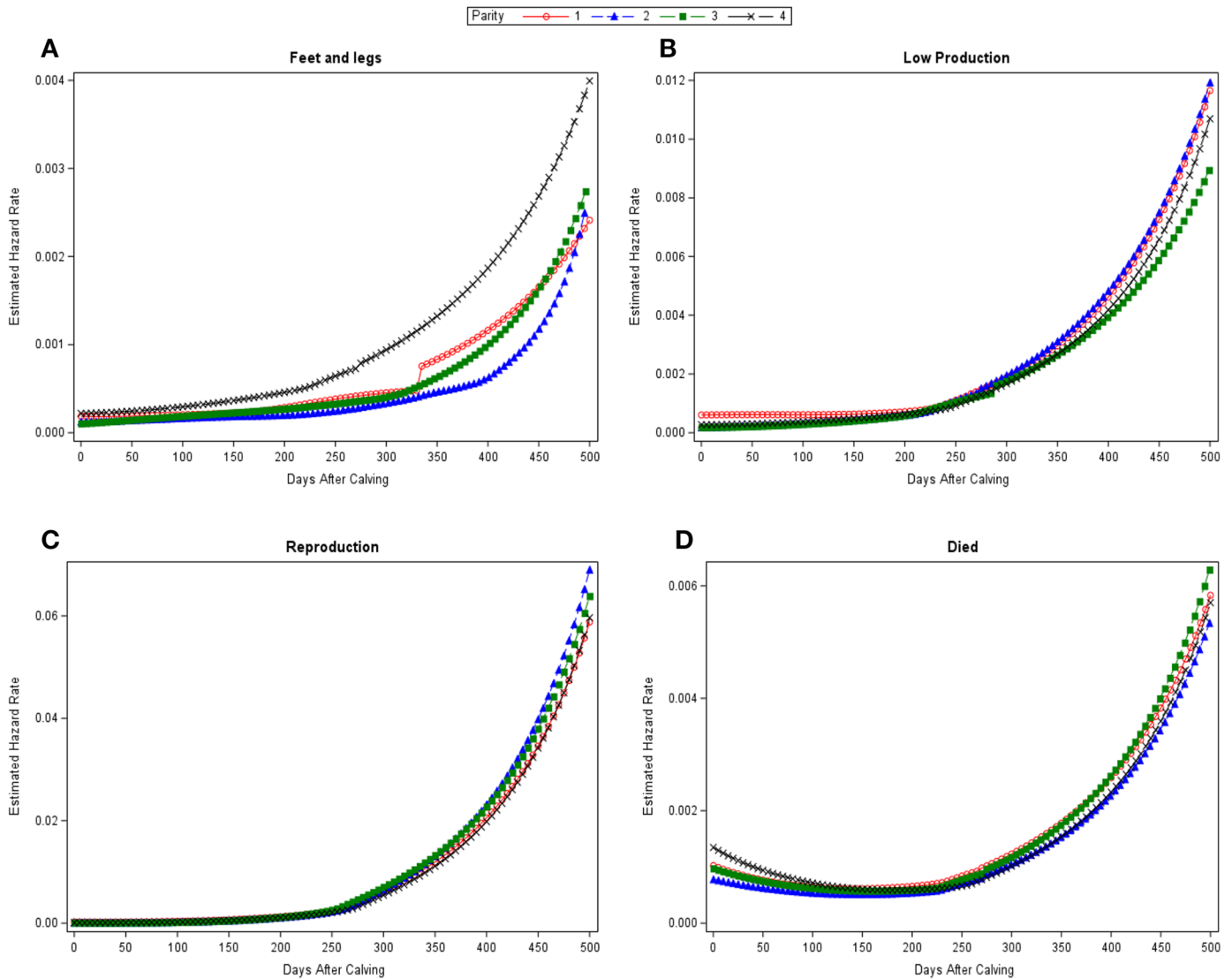


Figure 3. Distribution of culling risk ($\times 100,000^{-1}$) by days after calving and parity with culling reasons (A) feet and legs, (B) low production, (C) reproduction, and (D) died (scales vary for y-axes): parity 1 (red circles); parity 2 (blue triangles); parity 3 (green squares); parity ≥ 4 (black $\times$).

Monthly and geographic variations in disposal code frequencies are presented in Figure 2. A clear seasonal trend was observed for cows with culling code “reproduction” in the overall, northeast, and south regions, showing a noticeable decline during the winter season (February to April) and an increase following the summer (July). Although a similar decline was seen in the west during the winter, no consistent pattern was identified after the summer. The increase in the culling rate due to reproduction after summer may reflect challenges in achieving pregnancy during the hot months (Pinedo et al., 2010). De Vries and Risco (2005) also identified the effects of season on the reproductive performance with a significantly greater pregnancy rate in fall compared with the summer season. The south region had the highest propor-

tion of “reproduction” related culling (50%) compared with 25% in the northeast and 40% in the west. Previous studies similarly reported higher “reproduction” related culling in the southern region, potentially due to extended exposure to high levels of heat and humidity (Smith et al., 2000; Pinedo et al., 2010). Heat stress during the hot summer months is known to impair fertility in cows (Jordan, 2003), which may explain the regional differences in reproduction-related culling events (Vergara et al., 2014).

Hazard Distributions for Reason-Specific Culling

The risk for culling due to feet and legs, low production, reproduction, and died progressively increased from 300 d postcalving, peaking at 500 d in all parity groups

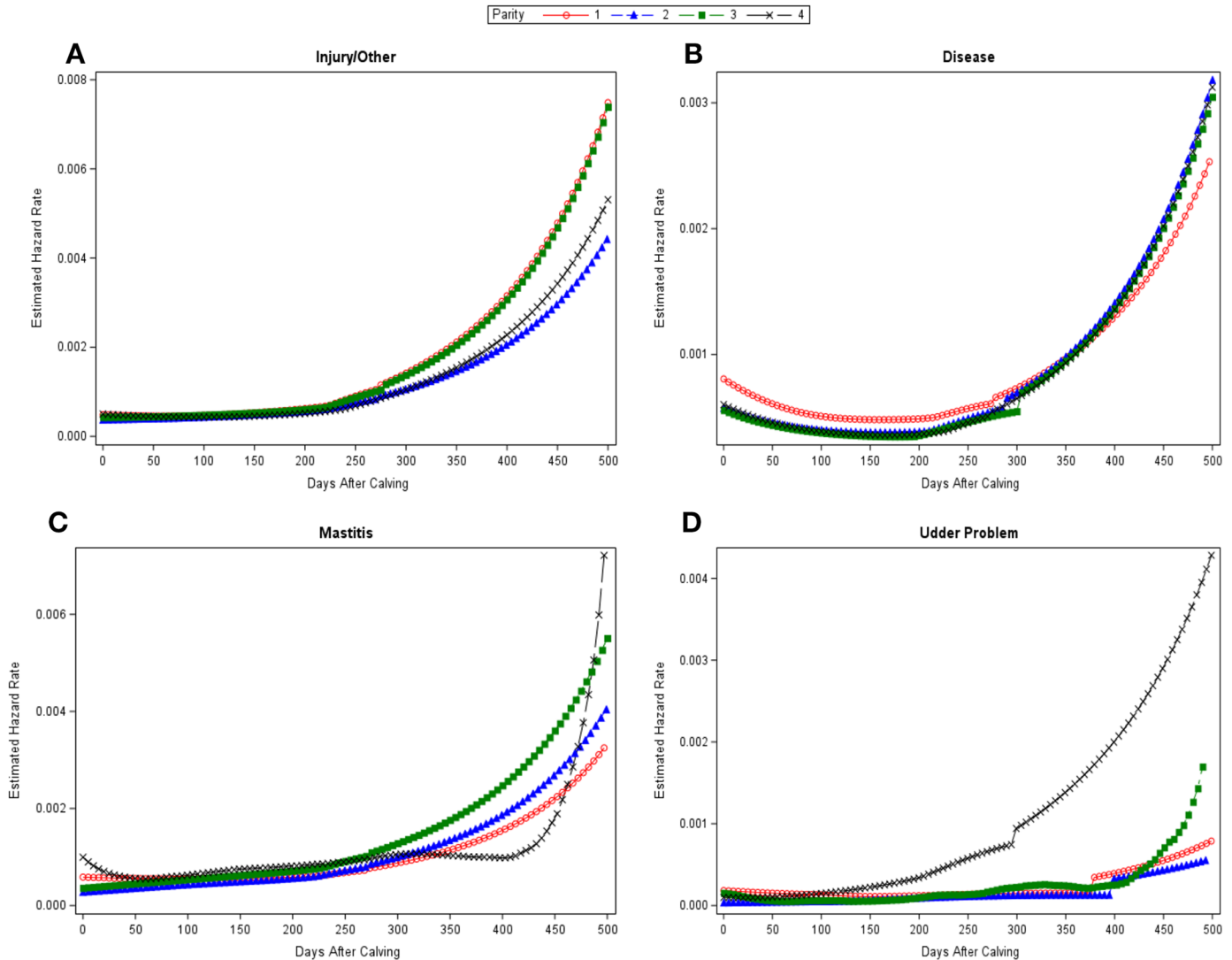


Figure 4. Distribution of culling risk ($\times 100,000^{-1}$) by days after calving and parity with culling reasons (A) injury/other, (B) mastitis, (C) disease, and (D) udder problem (scales vary for y-axes): parity 1 (red circles); parity 2 (blue triangles); parity 3 (green squares); parity ≥ 4 (black $\times$).

(1, 2, 3, ≥ 4 ; Figure 3). The risk for feet and legs as disposal code was higher in cows in parity ≥ 4 than in other-parity groups. The risk of culling due to low production was higher in first- and second-parity cows up to 150 d after calving and after 350 d postcalving than cows in third and fourth parities. The risk for died as the disposal code was higher in cows in parity ≥ 4 than in other-parity groups during the first 100 d postpartum.

Similarly, the risk of culling related to injury/other, mastitis, and disease as disposal codes began to increase from 300 d after calving, reaching a peak at 500 d across all parities (Figure 4). Cows in parity 1 experienced higher risk for injury after 300 d postcalving compared with other-parity cows. For early lactation (up to 50 d after calving) and late lactation (≥ 480 d after calving)

cows in parity ≥ 4 had higher risk of culling due to mastitis compared with cows in other parities. Susceptibility to IMI increases with age because older cows often experience concurrent health problems such as lameness and a decline in immune function (Breen et al., 2009). Disease-related culling in cows up to 300 d after calving was higher in primiparous cows than in multiparous cows, however, the risk was higher in multiparous cows after 300 d postcalving than the primiparous cows.

Udder-related culling was more prevalent in parity ≥ 4 cows than in other-parity groups. The increase in udder-related culling with parity may be attributed to physiological alterations in the mammary gland, with increase in size (Klaas et al., 2004) and changes in teat shape (Guarín and Ruegg, 2016), which can elevate the risk of

Table 3. Multivariate logistic regression for risk of death culling and live culling within 60 d after calving by multiple variables

Subgroup	Death			Live culling		
	OR ¹	95% CI	P-value	OR ¹	95% CI	P-value
Cow-relative 305ME milk yield						
<-1,944	0.67	0.20–2.2	0.52	1.4	0.86–2.41	0.15
–,1944 to 1,805	0.10	0.03–0.35	0.0002	0.29	0.17–0.48	<0.0001
>1,805 (referent)	—	—	—	—	—	—
Herd-relative 305ME milk yield						
<-2,118	<0.001	<0.001–0.002	<0.0001	<0.001	<0.001–< 0.001	<0.0001
–2,118 to 1,823	0.04	0.006–0.32	0.002	0.03	0.01–0.12	<0.0001
>1,823	—	—	—	—	—	—
Herd size						
≤100	0.38	0.27–0.52	<0.0001	0.59	0.47–0.74	<0.0001
>100 (referent)	—	—	—	—	—	—
Calving season						
Winter	0.62	0.51–0.75	<0.0001	0.94	0.83–1.04	0.21
Spring	0.70	0.57–0.87	0.001	0.93	0.82–1.03	0.17
Summer	0.78	0.64–0.95	0.01	0.76	0.68–0.85	<0.0001
Fall (referent)	—	—	—	—	—	—
Parity						
1 (referent)	—	—	—	—	—	—
2	1.20	0.98–1.48	0.06	1.38	1.24–1.53	<0.0001
3	1.08	0.88–1.32	0.44	1.27	1.14–1.42	<0.0001
≥4	0.78	0.65–0.92	0.005	1.03	0.94–1.13	0.45

¹OR = odds ratio.

teat injuries, milking machine failure, or leakage. These findings are consistent with those by Pinedo et al. (2014), who also reported low productivity, reproduction failure, and injury/other as contributors for culling in early and late lactation. In the transition period, because of a compromised immune system, most of the metabolic disease occurs, making the transition period a critical time for survival (Vergara et al., 2014; Horst et al., 2021). The reason low productivity and reproduction failure contribute to culling in advanced lactation is because cows are often diagnosed open during pregnancy check at dry-off (Pinedo et al., 2014) and the culling decisions based on reproduction and milk performance are typically implemented at the end of lactation, around the time of dry-off.

Logistic Regression Analyses

The odds of live and death culling within 60 d after calving across subgroups of variables of interests are presented in Table 3. Cow-level factors that were significantly ($P < 0.05$) associated with the risk of dying within 60 d after calving included calving season, parity, and cow-relative 305ME milk yield. Cows calving in the fall had higher odds of dying within 60 d postpartum compared with those calving in winter, spring, or summer, and the odds were greater in primiparous cows than in cows with parity ≥ 4 . The elevated early-lactation mortality in primiparous cows aligns with their higher culling rates compared with older cows. The odds of live culling within the first 60 DIM were higher for cows in pari-

ties 2 and 3 compared with first-parity cows. In contrast, Pinedo et al. (2014) and Dechow and Goodling (2008) reported the increment of live culling rate within 60 DIM with increase in parity. However, when considering the entire lactation, the annualized live culling rate was higher in first-parity cows than in second- and third-parity cows. The observed results suggest that although cows in second and third parity are more vulnerable for live culling within 60 DIM, primiparous cows are at greater risk over the full lactation period. Herd-relative 305ME milk yield and herd size were both significantly associated with the likelihood of dying within 60 d after calving. Cows in high-yielding herds had greater odds of death compared with cows in intermediate-yielding herds and low-yielding herds as corroborated by the result of high frequency of culling for cows in high-yielding herds due to feet and legs, reproduction issue, and death as shown in Table 1. In contrast to the present study, Pinedo et al. (2014) reported that cows in high-yielding herds had substantially lower odds of both death and live culling within 60 d after calving compared with cows in low-yielding herds when analyzing the lactation records from 16 conventional dairy herds in Texas. This discrepancy suggests that differences in production systems may influence the risk of culling across parities. The odds of died were higher in larger herds than in smaller herds (Table 3). Similarly, Pinedo et al. (2014) reported greater odds of death culling within 60 d after calving in large herds compared with smaller herds. As discussed previously, reduced opportunity for individual-cow monitoring in

Table 4. Multivariate logistic regression for risk of culling due to feet and legs, low production, and reproduction by multiple variables

Subgroup	Feet and legs			Low production			Reproduction		
	OR ¹	95% CI	P-value	OR ¹	95% CI	P-value	OR ¹	95% CI	P-value
Cow-relative 305ME milk yield									
<-1,944	0.73	0.59–0.89	0.002	3.43	2.96–3.98	<0.0001	0.48	0.43–0.54	<0.0001
-1,944 to 1,805	0.95	0.80–1.14	0.61	1.42	1.24–1.63	<0.0001	0.82	0.74–0.90	<0.0001
>1,805 (referent)	—	—	—	—	—	—	—	—	—
Herd-relative 305ME milk yield									
<-2,118	4.50	3.71–5.47	<0.0001	0.66	0.57–0.76	<0.0001	0.09	0.08–0.11	<0.0001
-2,118 to 1,823	1.32	1.10–1.60	0.003	0.92	0.81–1.06	0.28	0.80	0.72–0.88	<0.0001
>1,823 (referent)	—	—	—	—	—	—	—	—	—
Herd size									
≤100	0.83	0.69–1.01	0.06	1.13	0.96–1.32	0.11	0.78	0.67–0.90	0.001
>100 (referent)	—	—	—	—	—	—	—	—	—
Calving season									
Winter	0.93	0.85–1.02	0.19	1.06	0.99–1.14	0.06	0.91	0.87–0.96	0.001
Spring	1.02	0.93–1.11	0.65	1.02	0.95–1.09	0.49	0.95	0.90–1.00	0.07
Summer	0.95	0.87–1.03	0.24	0.99	0.93–1.06	0.95	1.07	1.02–1.13	0.003
Fall (referent)	—	—	—	—	—	—	—	—	—
Parity									
1 (referent)	—	—	—	—	—	—	—	—	—
2	1.11	1.00–1.23	0.03	0.91	0.85–0.97	0.008	0.91	0.86–0.96	0.001
3	1.23	1.11–1.37	<0.0001	0.76	0.71–0.82	<0.0001	0.82	0.77–0.86	<0.0001
≥4	1.84	1.68–2.01	<0.0001	0.73	0.68–0.77	<0.0001	0.67	0.64–0.70	<0.0001

¹OR = odds ratio.

larger herds may contribute to increased early-lactation mortality and removals (Shahid et al., 2015). These same factors, along with the calving season (fall vs. summer) were also significantly associated with the risk of live culling during the first 60 d postpartum.

The odds ratios for reason-specific culling risks are presented in Tables 4, 5, and 6. Cow-relative 305ME, herd-relative 305ME, and parity were linked to the likelihood of culling due to feet and legs, low production, and died. These same variables, along with herd size and season of calving, were associated with the culling due to reproduction, mastitis, and disease. Injury/other reason for culling was significantly related to herd-relative 305ME, season of calving, and parity, and udder problem was associated with cow-relative 305ME and parity.

Low-producing cows were at 27% lower risk of being culled due to reason feet and legs, 52% lower risk due to reproduction (Table 4), and 47% lower risk due to reason died (Table 6), compared with high-producing cows. Similar results of low culling risk for low-producing cows than high producers because of the reasons reproduction or died were reported by Pinedo et al. (2010). In contrast, low-producing cows had higher odds of being culled for low production, mastitis, and udder problem compared with the high-producing cows. This pattern also aligns with the Pinedo et al. (2010) findings, who reported the high risk of culling for low-producing cows with the reasons low production, mastitis, or udder problem.

For the reason feet and leg, cows in parities 2, 3, and ≥4 were 11%, 23%, and 84% more likely to be culled

than first-parity cows respectively (Table 4). For injury/other cause, parities 2 and 3 were both 18% more likely to be culled than first-parity cows (Table 5). For mastitis, cows in parities 2, 3, and ≥4 were 10%, 36%, and 44% more likely to be culled than first-parity cows. Pinedo et al. (2010) similarly reported increased culling rate with advancing parity for reasons like feet and legs and mastitis. Additionally, cows in parities 3 and ≥4 faced 1.3- and 2.7-times higher risk of culling due to udder problems, and 1.1- and 1.07-times higher risk due to died, compared with first-parity cows. This may be related to the higher milk production in older-parities cows, which increases the risk of udder problems and lameness, both of which are more prevalent in higher-parity cows.

However, for low production, reproduction, and disease, multiparous cows had a lower risk of culling compared with primiparous cows. Low-production cows in parities 2, 3, and ≥4 were at 9%, 24%, and 27% lower risk of culling, respectively, than, primiparous cows (Table 4). Similarly, for reproduction, cows in parities 2, 3, and ≥4 had a 9%, 18%, and 33% lower risk of culling than primiparous cows (Table 4). Regarding disease-related culling, cows in parities 3 and ≥4 both exhibited a 17% lower risk of being culled compared with primiparous cows (Table 5). These findings align with those of Pinedo et al. (2010) and Armengol and Fraile (2018), who reported decreasing culling rate with advancing parity for the reason low production, reproduction, and disease in conventional Holstein herds. Similarly, Portner and Heins (2024) reported higher culling rates of first-parity cows

Table 5. Multivariate logistic regression for risk of culling due to injury/other, mastitis, and disease by multiple variables

Subgroup	Injury/other			Mastitis			Disease		
	OR ¹	95% CI	P-value	OR ¹	95% CI	P-value	OR ¹	95% CI	P-value
Cow-relative 305ME milk yield									
<−1,944	1.07	0.91–1.25	0.39	2.27	1.92–2.69	<0.0001	0.76	0.62–0.94	0.01
−1,944 to 1,805	0.95	0.82–1.10	0.53	1.37	1.17–1.60	<0.0001	1.04	0.85–1.27	0.65
>1,805 (referent)	—	—	—	—	—	—	—	—	—
Herd-relative 305ME milk yield									
<−2,118	1.47	1.25–1.71	<0.0001	1.49	1.27–1.76	<0.0001	4.50	3.71–5.47	<0.0001
−2,118 to 1,823	1.09	0.94–1.26	0.21	1.37	1.17–1.60	<0.0001	1.32	1.10–1.60	0.003
>1,823 (referent)	—	—	—	—	—	—	—	—	—
Herd size									
≤100	1.04	0.86–1.27	0.63	1.23	1.05–1.44	0.008	0.76	0.59–0.97	0.02
>100 (referent)	—	—	—	—	—	—	—	—	—
Calving season									
Winter	0.94	0.87–1.01	0.10	1.11	1.04–1.18	0.001	1.03	0.95–1.11	0.45
Spring	0.89	0.83–0.96	0.003	1.03	0.97–1.11	0.25	1.07	0.98–1.16	0.09
Summer	0.83	0.78–0.89	<0.0001	1.01	0.95–1.08	0.62	1.0	0.92–1.08	1.00
Fall (referent)	—	—	—	—	—	—	—	—	—
Parity									
1 (referent)	—	—	—	—	—	—	—	—	—
2	1.18	1.10–1.27	<0.0001	1.10	1.03–1.19	0.005	0.95	0.88–1.03	0.31
3	1.18	1.09–1.27	<0.0001	1.36	1.27–1.46	<0.0001	0.83	0.76–0.91	<0.0001
≥4	0.94	0.88–1.01	0.12	1.44	1.35–1.53	<0.0001	0.83	0.77–0.89	<0.0001

¹OR = odds ratio.

due to reproduction and low production in organic herds compared with multiparous cows across multiple breeds (Holstein, Jersey, Montbéliarde, Normande, and Viking Red crossbred). Primiparous cows are expected to face a higher risk of culling for low production because they have not yet reached their peak physiological maturity; multiparous cows generally achieve significantly greater milk yields as they reach mature body size (Wathes et al., 2007) and therefore are retained longer in herds. Portner and Heins (2024) noted that because growing herds bring in more first-lactation cows, reproductive culling is more common in younger cows that fail to conceive. Ahlman et al. (2011) reported a higher culling rate of primiparous organic Swedish red cows compared with multiparous cows due to the disposal reason reproduction (low fertility) and low production. However, comparisons across studies should be interpreted cautiously because of differences in breed composition and production systems.

Age at Culling and Association with Genetic Merits of Animals

The average age of culling in this dataset of organic dairy herds was 62.4 ± 0.07 mo (5.2 years). In comparison, a study by De Vries (2017) analyzing data from 9,755 US Holstein conventional herds enrolled in the DHI program and processed by DRMS reported a shorter average longevity (the days from birth to culling) of 58.9 mo (4.9 years). This suggests that organic herds in the US tend to have a higher proportion of older cows than conventional herds.

However, the difference in longevity may partly reflect variation in the study period, so comparisons between organic and conventional systems should be interpreted with caution. Although conducted in only 3 states, a survey examining the herd health, management and characteristics across organic and conventional farms in New York, Wisconsin, and Oregon also found that organic farms had a greater proportion of older cows than conventional farms, with 14% versus 10% for third-parity cows, 8% versus 5% for fifth-parity cows, and 9% versus 5% for sixth-parity cows (Stiglbauer et al., 2013). The mean lactation number was also higher in organic herds (2.6 vs. 2.3 in conventional herds; Stiglbauer et al., 2013). The longer retention of cows in organic herds could be favored by their higher retention payoff value (Shrestha et al., 2025) and the higher cost of raising replacement animals in organic dairy (Tranel, 2020).

Globally, age at culling varies significantly by location; conventional herds in the Netherlands reported an average of 69.6 mo (Vredenberg et al., 2021), with 61.5 mo in Denmark (Thomsen and Houe, 2023), 81 mo in Australia (Workie et al., 2021), and 62.8 mo in Sweden (Owusu-Sekyere et al., 2023). These findings indicate that the average longevity of dairy cattle varies with production system and local management considerations.

We identified that the average productive lifespan (the time from first calving to culling) of 38.4 mo (3.2 yr), assuming an average at first calving of 24 mo (De Vries, 2020; Atashi et al., 2021). According to Norman (2022), the average productive life of Holstein cows in

Table 6. Multivariate logistic regression for risk of culling due to udder problem and died by multiple variables

Subgroup	Udder problem			Died		
	OR ¹	95% CI	P-value	OR ¹	95% CI	P-value
Cow-relative 305ME milk yield						
<-1,944	1.32	0.97–1.81	0.07	0.53	0.45–0.62	<0.0001
-1,944 to 1,805	1.33	1.00–1.77	0.04	0.89	0.77–1.03	0.13
>1,805 (referent)	—	—	—	—	—	—
Herd-relative 305ME milk yield						
<-2,118	1.09	0.82–1.46	0.53	3.44	2.97–3.99	<0.0001
-2,118 to 1,823	1.13	0.86–1.49	0.34	1.05	0.91–1.21	0.46
>1,823 (referent)	—	—	—	—	—	—
Herd size						
≤100	0.89	0.67–1.18	0.42	1.14	0.98–1.32	0.08
>100 (referent)	—	—	—	—	—	—
Calving season						
Winter	1.00	0.88–1.13	0.99	1.02	0.95–1.09	0.47
Spring	1.07	0.95–1.22	0.23	0.96	0.90–1.03	0.36
Summer	1.07	0.96–1.21	0.20	0.98	0.92–1.05	0.66
Fall (referent)	—	—	—	—	—	—
Parity						
1 (referent)	—	—	—	—	—	—
2	0.79	0.67–0.93	0.42	1.03	0.96–1.11	0.28
3	1.28	1.10–1.48	0.001	1.10	1.03–1.18	0.004
≥4	2.72	2.41–3.07	<0.0001	1.07	1.01–1.14	0.01

¹OR = odds ratio.

the United States is 28 mo. De Vries and Marcondes (2020) also identified the shorter average productive life of 33 mo in US conventional herds, which is lower than the value reported for organic cows in the current study. This difference could be attributed to the production system, with organic operations in the United States tending to retain cows longer than the conventional counterpart (Stiglbauer et al., 2013) due to higher costs of replacement animals (Tranel, 2020). A previous study by Shrestha et al. (2025) reported that retention pay off values were \$200 to \$1,000 higher in organic herds compared with their conventional counterparts, possibly benefiting extended cow retention in organic systems and resulting in longer productive life. Among organic dairy farms, Ahlman et al. (2011) also found an average of 38.4 mo of productive life for Swedish Holstein cows in organic herds. However, direct comparisons between US and European organic systems should be made cautiously because of differences in organic production standards (Ruegg, 2009).

We identified a moderate negative correlation between average age at culling and 4 major lifetime genetic merit indices, including net merit (−0.35), cheese merit (−0.35), fluid merit (−0.34), and grazing merit (−0.33). Although these indices include genetic predictions for productive life, their negative association with observed (phenotypic) age at culling should be interpreted cautiously because these are composite economic indices dominated by traits such as milk, fat, and protein yield, which receive substantially higher economic weights

than longevity-related traits. Consequently, correlations with overall index values may primarily reflect selection emphasis on production rather than true genetic merit for survival.

Nevertheless, the observed pattern indicates that realized longevity is strongly influenced in organic farms by environmental and management factors beyond genetic potential. These findings suggest that cows with higher genetic merit for economically important traits are more likely to be culled at a younger age. One possible reason is that genetically superior cows are often managed more intensively for productivity and reproductive performance, which may result in earlier culling decisions when performance expectations are not met (Weigel, 2006; Schuster et al., 2020; Shrestha et al., 2025). This highlights a potential trade-off between genetic selection for production efficiency and animal longevity, an issue that has gained attention in recent years because of its implications for animal welfare and sustainability (De Vries, 2017; Ali, 2021). Notably, although the EBVs of productive life have shown improvement over time, actual phenotypic measures of productive life have continued to decline (De Vries, 2017; Schuster et al., 2020; CDCB, 2025a). Our dataset did not contain EBVs for productive life; therefore, we were not able to use them for our analysis.

Another probable reason is that these merit indices are calculated using data primarily from conventional dairy system farms, and the reliability of underlying EBVs, particularly for productive life, can vary substantially

among animals, potentially contributing to discrepancies between genetic predictions and observed performance. Therefore, cows selected for high net merit may not perform as well or live as long under the different management, feeding, and health constraints of organic herds.

This study provides an overview of culling rates and patterns across the United States using various disposal codes. However, the findings may not fully represent individual herds in different states because of variations in herd size and farm-level management practices. Culling decisions often differ between small and large farms, and among organic herds, some are fully grazing-based whereas others graze their cows for only about 120 d per year to meet the NOP requirements.

CONCLUSIONS

Reproduction was the most reported reason for culling in organic dairy herds. The dynamics of culling risk with several reasons of culling was influenced by factors such as lactation stage, parity, milk production, herd-level characteristics, and calving season. Culling was more likely to occur during early and late stages of lactation. As days after calving increased, the risk of culling also increased for reasons including reproduction, low production, feet, and legs, injury/other, died, mastitis, udder problem, and disease. Shortly after calving, culling was mainly due to disease, udder problem, and died.

NOTES

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Nonstandard abbreviations used: 305ME = 305-d mature equivalent; DRMS = Dairy Records Management System; NOP = National Organic Program; OR = odds ratio.

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ORCIDS

- Bhuvan Shrestha,  <https://orcid.org/0000-0002-0512-8359>
Pablo J. Pinedo,  <https://orcid.org/0000-0001-7111-3377>
Rajesh Neupane,  <https://orcid.org/0000-0002-7009-4079>
Juan M. Piñeiro,  <https://orcid.org/0000-0002-5212-3817>
Sushil Paudyal  <https://orcid.org/0000-0002-6388-921X>